Secure wiring and finish the design

antocciphysics | TEACHER KEY | CPA

Learning target: Document permanent connections and evaluate added mass or friction.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 8, Day 3; 4 course days later.*
Ideal speed becomes 4 times as large. Actual rpm, load or slipping may change.

02. Explain a worked example (worked)
Clearly labeled functional features; actual attachment follows teacher-approved materials and procedure.

03. Look and explain (model)
Student-specific diagram with secure insulated connections and clear moving parts.

04. Build the solution (practice)
Added 50 g; acceleration decreases when net force is unchanged.

05. Use the idea (reason)
Rubbing adds resistance/frictional heating and may damage insulation; reroute and secure wire with power disconnected.

06. Choose a method (mixed) *Review: Unit 7, Day 13; 12 course days later.*
New field is 1/4 of the original under the long-wire model.

07. Exit check (exit)
Actual observation and response, not a generic completed box.

08. STRETCH: Apply it another way (stretch)
Examples exposed conductor near opposite terminal, loose joint, wire touching gear, unsecured battery; link each to short, disconnection, drag or movement.

Teaching note: Permanent electrical work still requires the classroom’s inspection and tool-safety checkpoints.

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01. Bring back an earlier idea (retrieve) *Review: Unit 8, Day 3; 4 course days later.*
Torque ratio 6; speed ratio 1/6. Assumes ideal gears with negligible losses.

02. Explain a worked example (worked)
Clearly labeled functional features; actual attachment follows teacher-approved materials and procedure.

03. Represent and interpret (model)
Student-specific diagram with secure insulated connections and clear moving parts.

04. Solve and justify (practice)
1.50 and 1.20 m/s²; 20% reduction.

05. Analyze the reasoning (reason)
Rubbing adds resistance/frictional heating and may damage insulation; reroute and secure wire with power disconnected.

06. Choose a method (mixed) *Review: Unit 7, Day 13; 12 course days later.*
$B_f/B_i = 4/6 \approx 0.6667$, using $B \propto I/r$.

07. Exit check (exit)
Traction, gearing, motor load, alignment, wheel inertia and electrical condition also affect performance.

08. STRETCH: Apply it another way (stretch)
Before 1.5 A, after 1 A; faulty connection dissipates 2 W. Heat and voltage drop may indicate a poor joint; disconnect before inspecting.

Teaching note: Permanent electrical work still requires the classroom’s inspection and tool-safety checkpoints.